

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. (5 Marks)

Objective

To efficiently match drivers with passengers, minimize passenger waiting time, reduce travel distance, and improve customer satisfaction.

Why Reinforcement Learning?

Reinforcement Learning is suitable because traffic conditions, driver availability, and passenger requests continuously change. The RL agent learns the best driver-passenger matching strategy from experience and improves over time.

Environment

Ride-sharing platform.

Agent

Ride-sharing matching system.

State Space

- Driver location
- Passenger request
- Traffic conditions
- Driver availability

Action Space

- Assign a driver
- Reject request
- Reassign nearby driver

Reward Function

- +20 for quick pickup
- +15 for customer satisfaction
- -10 for long waiting time
- -5 for unnecessary travel

Policy

Select the available driver that maximizes reward.

Learning Process

The RL agent observes the current state, assigns a driver, receives a reward based on service quality, updates its policy, and repeats the process. Over time, it learns better matching strategies.

RL Architecture

Passenger Request



Environment



RL Agent



Driver Assignment



Passenger Feedback



Reward



Policy Update

Expected Outcomes

- Reduced passenger waiting time
- Efficient driver utilization
- Improved service quality
- Higher customer satisfaction

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. (5 Marks)

Objective

To recommend personalized study materials that improve student learning performance.

Why Reinforcement Learning?

Students have different learning patterns. RL continuously learns from student interactions and recommends the most suitable study material.

State Space

- Student performance
- Completed lessons
- Quiz scores
- Learning history

Action Space

- Recommend video
- Recommend quiz
- Recommend notes

Reward Function

- +10 for course completion
- +8 for high quiz scores
- -5 for skipped lessons

Policy

Recommend the material with the highest expected learning improvement.

Exploration vs Exploitation

Exploration

- Recommend new learning materials
- Introduce new topics

Exploitation

- Recommend materials similar to previous successful recommendations
- Focus on weak subjects

Learning Process

The agent receives student feedback, updates recommendations, and gradually improves learning outcomes.

RL Architecture

Student Activity



Environment



RL Agent



Recommendation



Student Feedback



Reward



Policy Update

Expected Outcomes

- Personalized learning
- Better exam scores
- Higher student engagement

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. (5 Marks)

Objective

Clean the maximum area while avoiding obstacles and minimizing energy consumption.

Why Reinforcement Learning?

The robot learns efficient cleaning paths from previous experiences without requiring predefined rules.

Environment

House with rooms and obstacles.

Agent

Robot vacuum cleaner.

State Space

- Robot position
- Dirt level
- Battery level
- Obstacle locations

Action Space

- Move forward
- Turn left
- Turn right
- Clean

Reward Function

- +10 for cleaning
- -10 for collision
- -2 for unnecessary movement

Policy

Choose actions that maximize cleaning efficiency.

Value Function

The value function estimates the future reward of each state.

Higher value → Better cleaning path.

Lower value → Inefficient path.

Learning Process

The robot learns better routes after every cleaning session.

RL Architecture

Room Sensors



Environment



RL Agent



Movement



Cleaning Result



Reward



Value Update

Expected Outcomes

- Faster cleaning
- Reduced battery usage
- Better obstacle avoidance

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. (5 Marks)

Objective

Maximize trading profit while minimizing financial risk.

Why Reinforcement Learning?

The cryptocurrency market changes continuously. RL adapts trading decisions based on market conditions.

State Space

- Current price
- Previous price
- Trading volume
- Market trend

Action Space

- Buy
- Sell
- Hold

Reward Function

- +20 for profitable trades
- -20 for losses
- +5 for successful holding decisions

Impact of Poor Reward Design

- Excessive risk-taking
- Frequent unnecessary trades
- Poor investment performance

Reward Design Challenges

- Immediate profit may not indicate long-term success.
- Poor reward design may encourage risky trading.
- Rewards should balance profit and risk.

Learning Process

The agent continuously updates its trading policy based on previous trading outcomes.

RL Architecture

Market Data



Environment



RL Agent



Buy/Sell/Hold



Trading Result



Reward



Policy Update

Expected Outcomes

- Better trading decisions
- Increased long-term profit
- Reduced financial risk

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

(5 Marks)

Objective

Efficiently distribute electricity, reduce power wastage, and maintain a stable power supply.

Why Reinforcement Learning?

Electricity demand changes throughout the day. RL continuously learns from demand patterns and optimizes power distribution in real time.

State Space

- Electricity demand
- Battery level
- Renewable energy availability

Action Space

- Increase supply
- Reduce supply
- Store energy

Reward Function

- +20 for balanced distribution
- -15 for overload
- -10 for energy waste

Policy

Select actions that maximize efficiency and reliability.

Learning Process

The RL agent observes electricity demand, selects the best action, receives a reward, updates its policy, and continuously improves distribution decisions.

RL Architecture

Electricity Demand



Environment



RL Agent



Power Distribution



Reward



Policy Update

Expected Outcomes

- Reduced power wastage
- Better electricity distribution
- Improved grid stability
- Lower operational cost

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. (5 Marks)

Objective

Adjust medication dosage to improve patient health while minimizing side effects.

Why Reinforcement Learning?

Patient conditions change over time. RL learns the best dosage by continuously analyzing patient responses.

State Space

- Patient health condition
- Blood pressure
- Heart rate
- Previous dosage

Action Space

- Increase dosage
- Reduce dosage
- Maintain dosage

Reward Function

- +20 for improved health
- -20 for adverse effects
- -10 for incorrect dosage

Learning Process

The agent observes patient health, recommends a dosage, receives feedback, updates its policy, and improves future recommendations.

RL Architecture

Patient Data



Environment



RL Agent



Medication Dosage



Patient Response



Reward



Policy Update

Expected Outcomes

- Better patient recovery
- Reduced medication errors
- Personalized treatment
- Improved healthcare quality

**7. Design a model drone delivery system uses Reinforcement Learning for navigation.
Recall how exploration and exploitation strategies influence route optimization.
(5 Marks)**

Objective

Deliver packages safely while minimizing travel time and energy consumption.

Why Reinforcement Learning?

Weather, obstacles, and traffic conditions change continuously. RL learns the most efficient delivery routes.

State Space

- Drone location
- Battery level
- Weather
- Obstacles

Action Space

- Move north
- Move south
- Move east
- Move west
- Land

Exploration vs. Exploitation

Exploration

- Try new routes.
- Learn alternative paths.

Exploitation

- Use previously successful routes.
- Minimize delivery time.

Learning Process

The drone learns from every delivery and gradually selects more efficient routes.

RL Architecture

Navigation Data



Environment



RL Agent



Route Selection



Delivery Result



Reward



Policy Update

Expected Outcomes

- Faster deliveries
- Lower battery usage
- Improved navigation
- Safer drone operations

8. Design and develop a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. (5 Marks)

Objective

Improve assembly line efficiency while reducing production time and errors.

Why Reinforcement Learning?

The robot learns the best sequence of assembly operations through continuous interaction with the production environment.

State Space

- Robot position
- Part availability
- Assembly status

Action Space

- Pick part
- Assemble
- Move
- Inspect

Reward Function

- +15 for successful assembly
- -10 for defective assembly
- -5 for delays

Policy

Select assembly actions that maximize productivity.

Value Function

The value function estimates the long-term reward of each assembly state, helping the robot choose the most efficient sequence of operations.

Learning Process

The robot updates its value estimates after each assembly task and continuously improves its performance.

RL Architecture

Assembly Data



Environment



RL Agent



Assembly Action



Production Result



Reward



Value Update

Expected Outcomes

- Faster production
- Higher product quality
- Reduced manufacturing cost
- Improved efficiency

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. (5 Marks)

Objective

Maximize click-through rate (CTR) and advertising revenue by showing relevant advertisements.

Why Reinforcement Learning?

User interests change over time. RL continuously learns which advertisements are most effective.

RL Components

Component	Description
Environment	Online advertising platform
Agent	Advertisement recommendation system
Policy	Display the advertisement with the highest expected reward

State Space

- User profile
- Browsing history
- Previous clicks

Action Space

- Show Advertisement A
- Show Advertisement B
- Show Advertisement C

Reward Function

- +10 for click
- +20 for purchase
- -5 for ignored advertisement

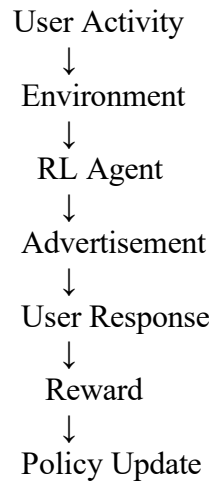
Reward Design Challenges

- Users may click accidentally.
- Purchases may occur much later than the advertisement.
- Poor reward design can lead to irrelevant advertisements.

Learning Process

The agent updates its recommendation policy after observing user interactions.

RL Architecture



Expected Outcomes

- Higher click-through rate
- Better user engagement
- Increased advertising revenue
- More personalized advertisements

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. (5 Marks)

Objective

Optimize water usage while maintaining healthy crop growth.

Why Reinforcement Learning?

Soil moisture, weather, and crop conditions change continuously. RL learns the optimal irrigation schedule from experience.

RL Components

Component	Description
Environment	Agricultural field
Agent	Smart irrigation controller

State Space

- Soil moisture
- Weather
- Temperature
- Crop growth stage

Action Space

- Increase watering
- Reduce watering
- Stop watering

Reward Function

- +20 for healthy crops
- +15 for water conservation
- -10 for water wastage
- -15 for insufficient irrigation

Policy

Choose the irrigation action that maximizes crop growth while minimizing water usage.

Learning Process

The RL agent observes field conditions, decides the irrigation level, receives rewards based on crop health and water usage, and updates its policy over time.

RL Architecture

Field Sensors



Environment



RL Agent



Irrigation Action



Crop Response



Reward



Policy Update

Expected Outcomes

- Efficient water usage
- Increased crop yield
- Reduced irrigation cost
- Sustainable farming practices

Code of Conduct:

*I, U. Somesh Kumar (192425019) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: U. Somesh Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand